

*To:* The Australian Senate Committee on Education and Employment Legislation

*From:* Max Yong, Harvard Kennedy School; Honorary Fellow at the Melbourne University Centre for the Studies of Higher Education

*Subject:* The budget-neutral way to fix Job-ready Graduates

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### **Executive Summary**

I offer a budget-neutral solution to the multiple policy failures of the Job-ready Graduates (JRG) package. My research, which was cited by the Australian Universities Accord and The Productivity Commission, shows that only around 1.5% of students changed their course preferences in response to JRG. Because students are highly price inelastic to fee changes, it is incumbent on the Australian Government to set fees that are fair across all fields of study. To help solve this, I develop a multi-objective optimization model to correct these distortions while maintaining current funding levels and accounting for political feasibility.

### **About the author**

Max Yong is an economist and policy researcher, currently studying a Master in Public Policy at Harvard University as a John Monash Scholar. He works as a Teaching Fellow in the Harvard University Department of Economics with Professor John Y. Campbell and Professor Dani Rodrik. Also, an Honorary Fellow at the Melbourne University Centre for the Study of Higher Education, Max conducted the first quantitative analysis of Job-ready Graduates in 2022 to show that the policy did not reach its intended effect of shifting students between different fields of study. Max is a contributor at *The Age* and *The Sydney Morning Herald*.

## **Background on Job-ready Graduates (JRG)**

In 2020, the Morrison Government dramatically changed undergraduate university fees across different fields of study. While some degrees became much cheaper – including teaching, nursing, and maths – others saw large fees hikes; most notably a 113% increase in fees for arts and humanities students.

## **The main problems of JRG**

There were three major problems:

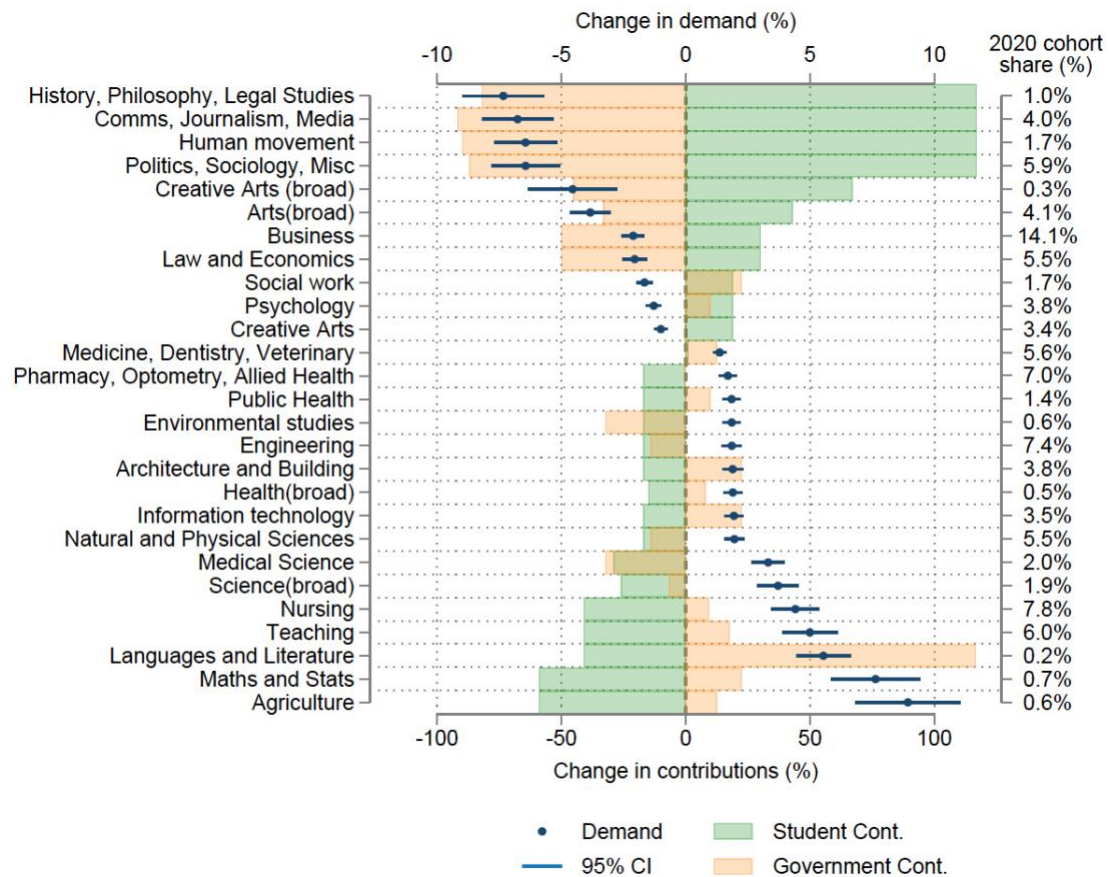
- 1. Due to HECS-HELP income contingent loans, students are highly price *inelastic*** (i.e., they barely respond to price changes)
- 2. The supply and demand incentives were misaligned**
- 3. Some courses are now very expensive, and the changes resulted in higher cost increases for women**

In more detail:

- 1. Due to HECS-HELP income contingent loans, students are highly price *inelastic*:**

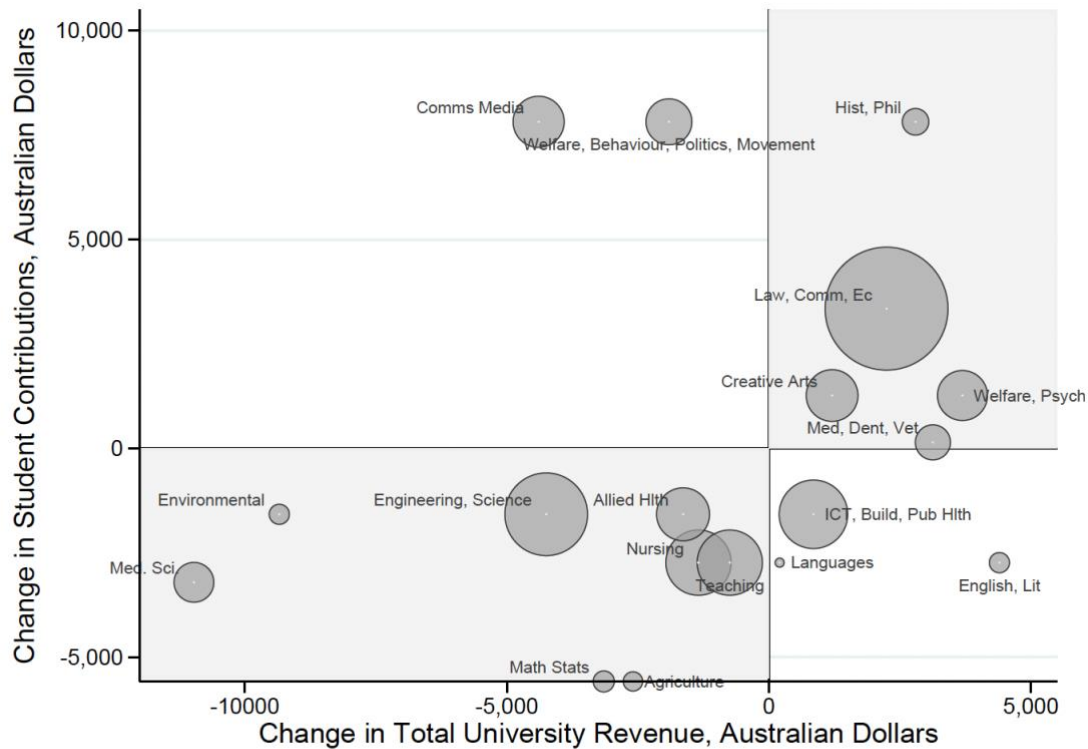
The stated objective of JRG was to divert university applicants into fields of study that would arguably support the needs of the future labour market. Areas of ‘national priority’ including nursing, agriculture and teaching were targeted with reduced fees. In contrast, supposedly ‘low need’ areas including arts, communications and law were targeted with fee increases.

However, research conducted by myself and coauthors at Melbourne University (and cited by the Australian Universities Accord and The Productivity Commission) showed that students were highly inelastic in response to the fee changes<sup>i</sup>. Despite very large fee changes, only around 1.5% of all students changed their preferences in response to the reforms.



**2. The supply and demand incentives were misaligned:** For many courses where the government was aiming to increase enrolment (incl. Nursing and Teaching) it reduced the amount of funding universities received per-student in these fields. It sent a positive demand signal, but a negative supply signal. In the below figure, this is represented in the bottom-right quadrant shaded grey.

They also did the opposite: sent a negative demand signal with higher prices for Creative Arts, History, Law, Commerce, etc. while sending a positive supply signal with higher university revenue-per-student in these same fields (represented in the top-right quadrant shaded grey).



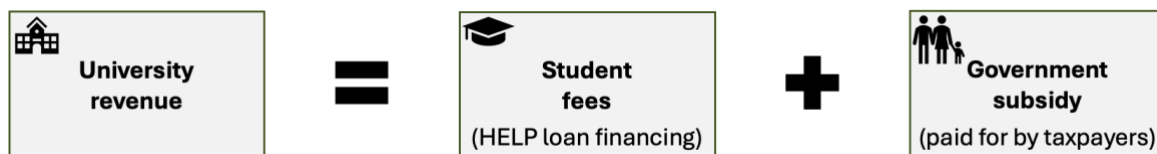
Notes: Circle sizes represent Bachelor Degree graduates aged 20 to 29 by aggregated field of study from the 2021 Australian Census (via ABS TableBuilder).

3. **Some courses are now very expensive, and the changes resulted in higher cost increases for women:** Under JRG, the maximum student contribution amount increased 28%. The most expensive three-year Bachelor courses are \$52,200 in 2026, which are in the fields of Communications, Society and Culture, Law, Accounting, Administration, Economics, Commerce<sup>ii</sup>. It is worth noting that expensive degrees may never be paid back by graduates (e.g., if they exit the workforce to become a full-time stay-at-home parent), which effectively results in a taxpayer-funded subsidy.

Women are worse off post-JRG, as their average yearly fees rose \$107.40 more than men<sup>iii</sup>.

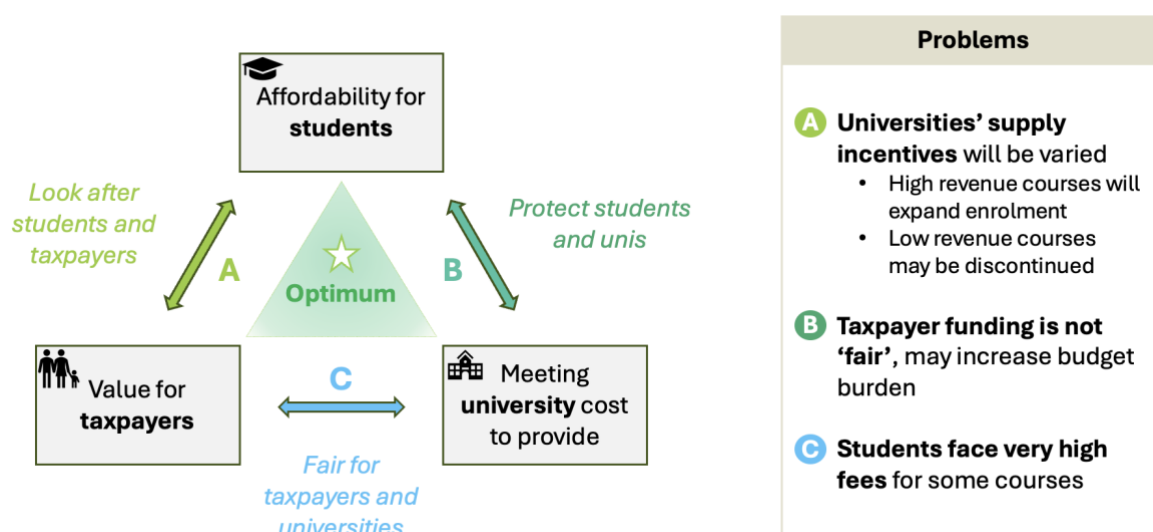
### **Why it is so hard to 'fix': The policy trilemma**

Students and universities are 'price-takers', and the government is the 'price-maker'. For each field of study, the Australian Government sets the student tuition and government subsidy amounts, which determines the amount of revenue universities receive per-student:



If the government wants to set two of these components, the third will be automatically determined. **This is a policy ‘trilemma’, where only two of three preferred outcomes can be chosen.**

Because students receive concessional income-contingent loans to fund their studies, they are relatively price insensitive, however student debt is still a burden in the future<sup>iv,v</sup>. University revenue amounts determine universities’ incentives to supply different courses, and government subsidy amounts come at direct cost to the taxpayer<sup>vi</sup>. This ‘trilemma’ means policymakers are unable to set fees at an optimal level for all stakeholders, thus trade-offs must be made.



Because the Albanese Government has announced a preference to fund more places in coming years, it is vital to find a sustainable funding model<sup>vii</sup>.

### **The solution: Optimizing across key criteria**

I argue that it is important to optimize the following criteria (in no explicit order):

- 1. Budget burden:** Minimize cost to taxpayers
- 2. Cost to students and repayment feasibility:** Minimize costs to students and ensure they can feasibly repay their HELP loans over the course of their career
- 3. University revenue:** Ensure funding amounts are adequate for universities to deliver courses at a high level of quality

4. **Alignment with labour market needs:** Ensure universities are producing enough graduates in each field of study to service the labour market
5. **Political achievability:** Acknowledge and work within political constraints (e.g., budget neutrality; no fee increases for nurses, etc.). *N.B. This point would not be relevant in an 'ideal' world, however, if political barriers are the reason that fee reform is blocked, they should be taken into account when strategically formulating a new policy for university fees*

I have developed a multi-objective model that optimises fees and subsidies subject to practical constraints. It is currently calibrated along the following principles:

- I. **Reduce the largest and least defensible JRG fee hikes.** That means cutting the very high student contributions imposed on fields such as humanities, communications and some society-and-culture disciplines
- II. **Keep aggregate government cost broadly neutral.** In the current fiscal environment, it is sensible to strive for budget neutrality in higher education policy reform
- III. **Minimize fee increases where possible.** A feasible reform cannot simply shift all the burden onto a few other fields. In my modelling work, applying political caps to fee increases is not incidental; it is central.
- IV. **Avoid major reductions in per-student university revenue.** If universities face large revenue losses in targeted fields, the supply response will again undercut the policy.
- V. **Protect politically and socially sensitive low-fee fields** such as teaching and nursing from fee increases, or at minimum hold them to a stricter cap than other disciplines

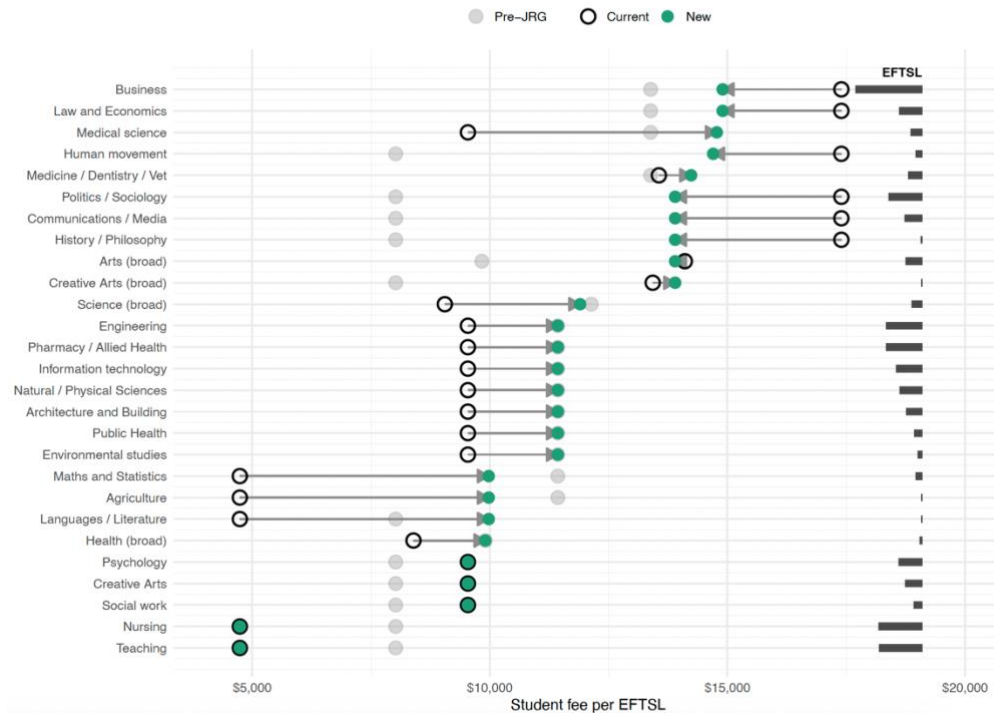
This pragmatic optimization approach recognises that reform is not a search for a single 'technically correct' fee schedule. It is a search for the best achievable schedule under fiscal and political constraints.

Based on these objectives and constraints, see three potential archetypes of outcomes:

- A. Rebalance student fees to improve fairness across disciplines, while staying budget neutral:** In this budget-neutral model, universities are paid the exact same as today. There is a rebalancing of fees across fields of study (the maximum fee would now only be \$14,900 per year). Any fee increases are strictly less or equal to pre-JRG levels. Important fields of study like nursing and teaching are unchanged.

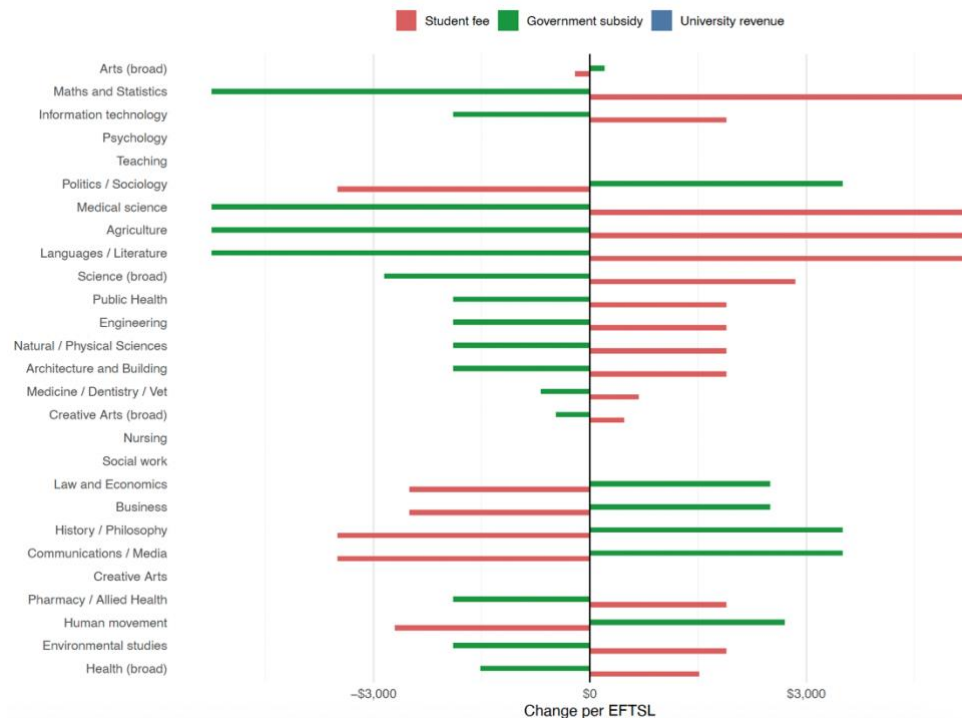
Current vs New Fee Schedule

Grey marker is the indexed pre-JRG fee. Right-side bars show EFTSL. Arrowheads appear only where fees change.



Funding Changes by Field

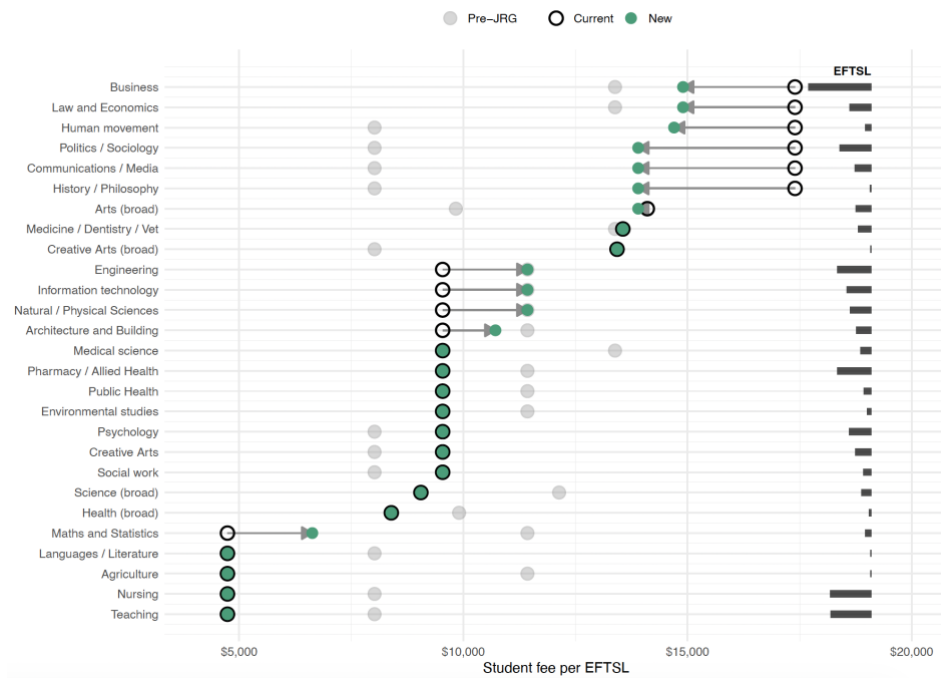
Per-EFTSL changes in student fees, government subsidy, and university revenue.



**B. Readjust per-student revenue to universities, while staying budget neutral:** In this budget-neutral model, per-student revenue to universities is adjusted closer to pre-JRG levels. This allows for a fairer rebalancing of fees. To avoid an overnight decrease in university revenue, the government could freeze government contribution amounts for a few years and allow inflation to create a real decrease over this period.

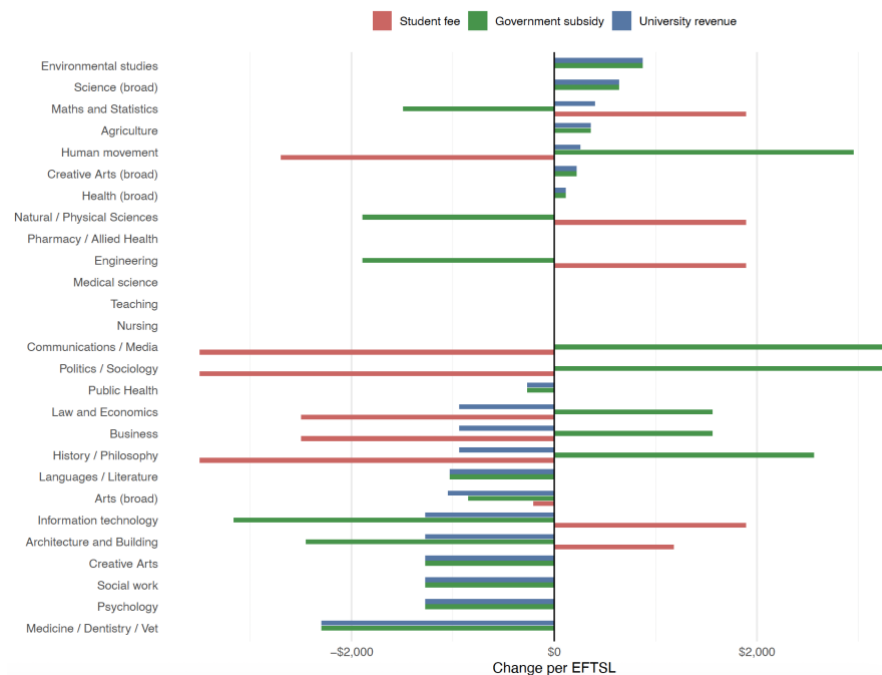
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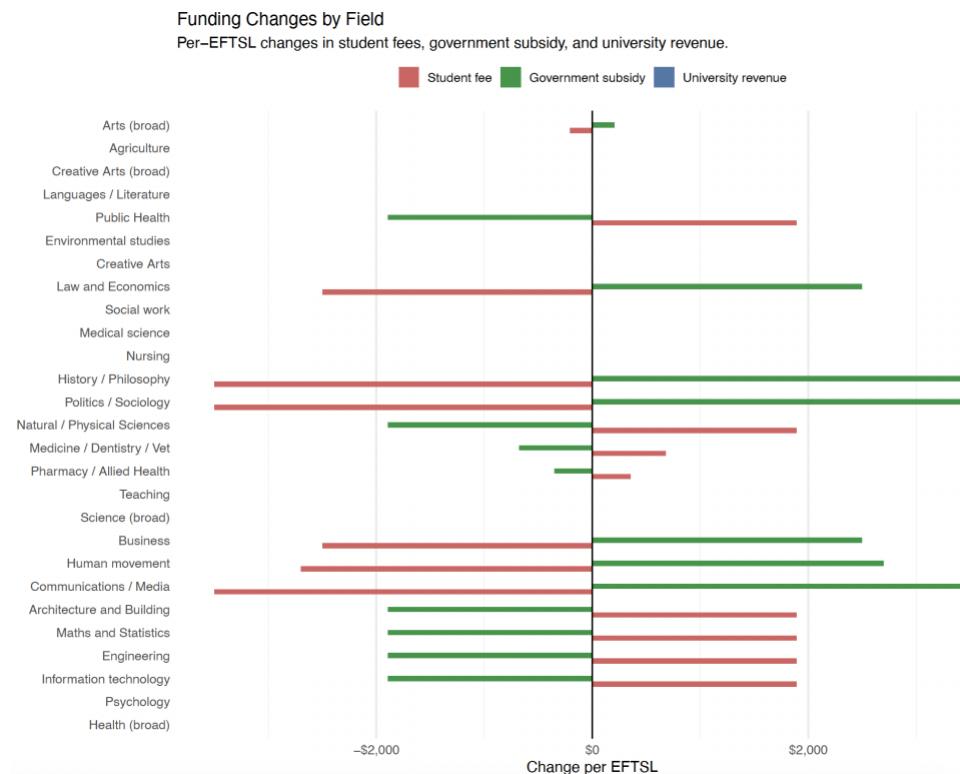
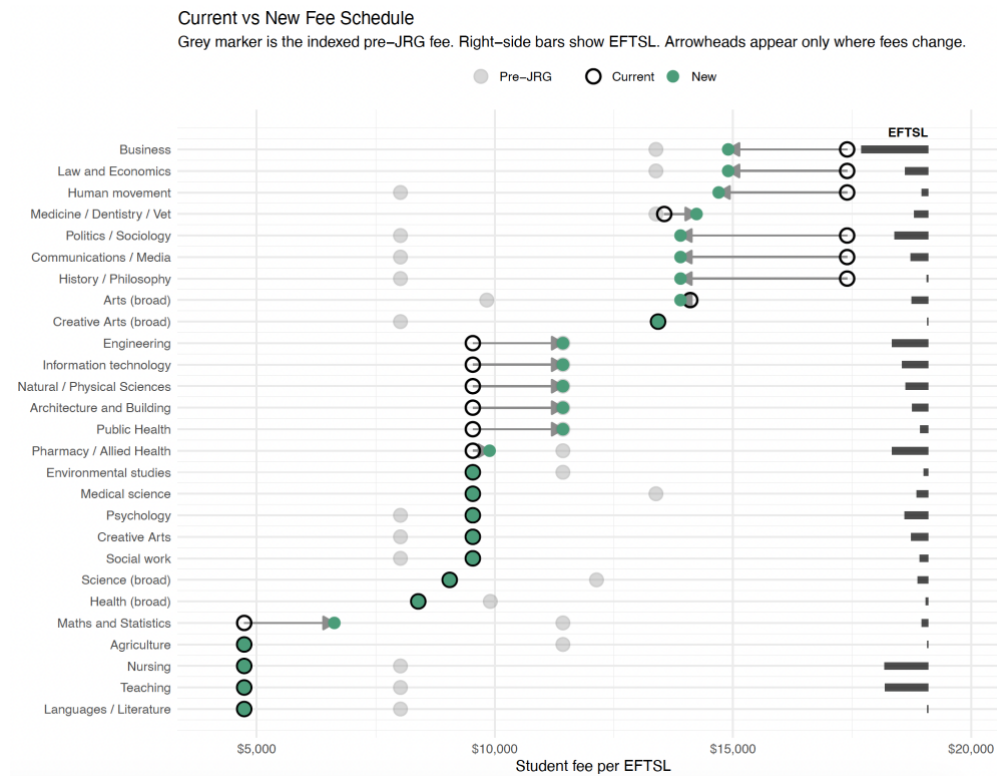
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Per-EFTSL changes in student fees, government subsidy, and university revenue.





- C. If government is willing to expand funding by only 3%, it can result in a fairer allocation for students and universities:** This non-budget neutral model results in highly favourable outcomes for students while keeping university funding unchanged. There are only a few small fee increases, with the remainder of fees decreased or unchanged. However, this would require higher budget outlay.



### **From here: Next steps to fix JRG**

While I propose one solution to fix JRG, I note that the model I have created can (and should) be tweaked depending on different preferences and inputs. I recommend that the Education Minister, the Department of Education (including ATEC) use these principles in conjunction with ongoing costing work to further refine the model.

However, I do caution that taking too long to solve a ‘technically’ correct answer will be unfair to current students who are paying as much as \$52,200 for a 3-year bachelor degree today. Costing for undergraduate degrees has long proven very difficult to estimate. So, I encourage the government to use its current political capital to move quickly and make university fees fair for students within the next year.

I am, of course, open to supporting this effort in any way I can.

## References

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- <sup>i</sup> <https://melbourneinstitute.unimelb.edu.au/publications/working-papers/search/result?paper=4751741>
- <sup>ii</sup> <https://www.studyassist.gov.au/financial-and-study-support/commonwealth-supported-places/student-contribution-amounts>
- <sup>iii</sup> Yong, M. (2022). Supply and demand effects of university funding changes: An Australian policy analysis. Honours thesis.
- <sup>iv</sup> <https://www.education.gov.au/higher-education-loan-program>
- <sup>v</sup> [https://melbourneinstitute.unimelb.edu.au/\\_\\_data/assets/pdf\\_file/0011/4751741/wp2023n11.pdf](https://melbourneinstitute.unimelb.edu.au/__data/assets/pdf_file/0011/4751741/wp2023n11.pdf)
- <sup>vi</sup> [https://www.econstor.eu/bitstream/10419/325009/1/cesifo1\\_wp12018.pdf](https://www.econstor.eu/bitstream/10419/325009/1/cesifo1_wp12018.pdf)
- <sup>vii</sup> <https://ministers.education.gov.au/clare/more-australians-get-crack-uni-2026-extra-9500-domestic-places-across-country>